

Inês Hipólito: The free energy principle in the edge of chaos

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and that i can say that it's my great great pleasure to introduce today's speaker inescipulito here in our ccnv seminar series at the free university in berlin where we have in this semester as a topic of the seminar series like tutorials on the free energy principle and predictive coding and some what was it half a year ago inescipulita was also organizing here big workshops that probably many people have attended on the free energy principle here in berlin over the berlin school of mind and brain so ines is affiliated to the ucl in london she is also a fellow at developmental psychology at the faculty of the behavioral science faculty at the university of amsterdam and is here in burden as lecturer at the mind and brain school which belongs to humboldt university and is dealing in her work as i understand it mostly with free energy principle uh uh in the context of interpreting the mind's function we had multiple talks in the series before which were introducing mostly technical aspects of free energy principle and computations within that we are very much looking forward to your talk today with this i think i directly hand over nice to have you thank you so much tim mohanda thank you so much for uh inviting me um joining uh these are wonderful um speakers that i've been having in this really great initiative group on the financial principle in in berlin um thanks um so much for the very kind introduction i am going to now try to share my slides hopefully uh there won't be any thing rent access in systems preferences yes i think that's fine yeah so the title of my talk is the financial principle on the edge of chaos i would like to just preface it uh by saying that um that uh it is a little bit of a talk on um the sorry on the the philosophy of science of the french principle and that's sort of like speaking to and relating to uh the most recent uh events um on that have been happening on the last year so if you could just give me the next slide perfect so yeah here we go um there's been a little bit of a discussion uh going on on the literature on the philosophy side um people are um talking to each other about this realism non-realism uh kind of approach um to the french principle of course in philosophy this is not

news um it is a very uh live discussion in philosophy of science in general about theories models um etc in science in scientific practice and more recently this uh debate has been revived in the light of the french principle and that's what i wanted to talk about today so if you go to the next slide yeah so um the lineup for today is just these three main points uh i will go very quickly over the realism debate and then i will introduce just the essentials of the free energy principle which of course people that are uh usual usually coming to these um to these seminars of course uh they're very very much acquainted but i'm going to highlight the points that we need to highlight in order to then come back and think about the free energy principle in the light of the realism debate okay so um next slide yes so we go now to the realism debate and uh nobody would be happy with this uh definition that i have here because uh the realism debate is so nuanced uh philosophically speaking that it will be it is very difficult to come up with a definition that will make everybody happy but let's try um so we can say about realism try to define it as uh the properties of a model are expected as somewhat an ontological property of the phenomenon of scientific interest so you should find some kind of isomorphism some kind of match between the properties of the model and the properties of the phenomenon under study the next slide so for example the property of study is as we are here precisely to study all of that things that we love the brain cognition or even cell activity next slide yes so basically now there are three or i was thinking about how can we uh put this together the literature is so vast and so integrated i was thinking okay let's try to put together at least three main groups or three main ways that we can think of uh the realism so the first one is uh the sort of like the more traditional one which is thinking about under realism thinking about models as accurate representations right and this comes for example from uh go to freight in sense and reference yeah um so the second one is um a little bit more complex uh but it's sort of like what we have um we find a lot in uh cognitive science or coming to neuroscience which is uh the idea that if models are obstructions and if they succeed to predict then the properties of the model should be literally present in the system being modeled it was was there a question oh i thought i had um a microphone okay uh so you find for example that in uh particularly uh very explicitly in uh this paper by michael briscolora defending a realist perspective on bayesian cognitive science the third way of being realist um in scientific practice is to claim um that claims made about the system of interest of scientific interest interest by truth preservation can also apply to the system itself in which case a target system is health to the properties of the model so this is found for example in these two papers that i have here which are very recent in the literature on the free energy principle and active

inference so that would be one way to connect or think about models made under active inference or free energy principle uh that also hold a realist perspective um on on the target system explained under the free energy or active inference oh yeah so then um what happens is to put it in a little bit more of a simpler way is that the mappings between the model and the target so the mappings between the model and the target allow to convert truths found in the model into claims about the target system so that's the very very basic idea that is being held in the this third realist assumption and then we have the converse or the counterpart antirealism and roughly the definition goes something like this the model is simply an instrumental tool that once applied to some activity of in scientific interest then the model allows to learn about and draw predictions about a system of scientific interest so the main gist is that as opposed to having an accurate match or isomorphism by virtue of this or that way of thinking or arguing for it the model is simply an instrumental tool that is an opportunity for learning next slide i think okay however the system under study does not need to have the properties of the model as of course it is argued for uh in the realist take on it right so that's why live debate going on next slide okay so now also a couple of ways of uh being anti-realist right so the first one is the meaningless way of thinking about models models are in themselves meaningless they do not hold semantic value other than that that is constructed under or by using them to understand or predict behavior and this was for example held by uh david hilbert or um boss van frassen next slide oh yeah um in the review models need not to be more than empirically adequate because they are mathematics um because mathematics is arithmetic and algebraic qualifier free identities so in this sense models themselves are semantic they do not hold any semantic value and they are qualifier free yep next slide okay then uh very much in agreement with um with uh david hilbert and uh blasphemy is albert einstein in this amazing lecture that he gave in berlin in 1921 if you hadn't have the opportunity to read it please do so and it tells us this was in berlin and it tells us as far as the laws of mathematics refer to reality they are not certain and as far as they're certain they do not refer to the reality it is certain that mathematics generally and particularly geometry owes its existence to the need which was felt of learning something about relations of real things to one another so of course this is very much in line with the anti-realism health by um david hilbert um and van vlassen next and i think i have something yeah and this is actually something that if you are interested on um i've just alluded to in the paper is just out in segmentary on michael miguel aguileras paper in physics of live reviews and we do go on and a little bit on that casper husband eye okay next slide anti-realism take two is that there is found as uh models thought of as idealizations or fictions uh in

science and the idea is that models allow us to abstract certain features of a phenomenon and they are for that reason by virtue of abstraction and opportunity for learning then take three on anti-realism is the view that the success of a model justifies the acceptance of the epistemic value of the model without necessarily making ontological claims so basically what we are saying is that the virtue of the model resides in its epistemic value and not in its ontological commitments or ontological commitments that one would make by virtue of using a model uh i think there's more on that slide yeah the success of our best scientific models um informed models um even if our model is extremely successful which of course we never know because nobody would ever claim that they have a model that is true in the best case scenario we would claim that we have a model that is viable so even if we have a very viable model that does not according to this uh view and andrew list view that does not allow us to um hold um a belief that everything in the model is true so not even that the model is viable oh yeah so then we find these ideas for example as far back as aussie under in his introduction to um copernicus magnus oppos and it says something like this the astronomer will take as their first choice that hypothesis which is the easiest to grasp the philosopher will perhaps rather seek the resemblance of the truth but neither of them will understand or state anything certain unless it has been divinely revealed to them now since one cannot in any way attain the true causes one will adopt whatever suppositions enable the motions to be computed correctly okay so this is take three the models are it's about the viability of the models rather than the truthiness of the models okay so next slide oh yeah so then um one last thought on anti-realism um take three is that then the hypothesis and assumptions that may or may not be encoded in a computational model are not intended to describe the way the object being modeled is actually structured but next simply serve to represent the observed data in an economical fashion and this is quite interesting because these are thoughts that weren't present quite quite a while a while back in time okay next slide so this was the setup uh of course he did not do any justice um to the realism debate which is absolutely rich super nuanced but at least you got an idea of like three takes of how to be realist about models um or uh three ways of being non-realist or instrumentalist if you will about models so now i want to say something about the fep just highlighting points that i find interesting in order to then come back to the realism debate and think about the fep under that realism debate and see what we make of it okay so then next slide all right so i want to say here that i take um typically i take my take on the fep is um so there are these two ways that you can get to the fep one is through the low road which you start with helmholtz um unconscious machine and then you work your way through vision

inference until it gets to active inference until you get to fep my take on my preferable take is the high road or what carl is called the high road which is where you start with biology and with certain observations that you make in the behavior of the world and then you put forward a sort of like first premise um about as an attempt to start explaining how systems living systems adaptive systems behave in the world in a way that that first premise um explains uh throughout across the board um adaptive behavior so this is my preferred um way i think it's though and that is most plausible and most interesting to me and it starts off with the question of how is it that living organisms persist while engaging in adaptive exchanges with the environment and this is an extraordinary thing because that's what i'm really obsessed about and i really like is that it's so extraordinary that you have an environment that is constantly changing so the only rule is that things change um and yet living systems do not succumb to those changes they actually persist in this changing continuously changing environment and this is the thought that i find extremely important because we should not lose sight of that thought because once we do that will get us into um other parts that i find at least i find less interesting such as minds as computers okay having said that um next slide so this is living systems maintain themselves by and then we continue maybe we can just open the yes by remaining in non-equilibrium steady states right they are open systems which is why as we all know the non-equilibrium steady states and s next one and they do that by restricting themselves to a limited number of states and why is that well because there is only a very a finite number of states that will keep them alive and um they will not um attain a certain face boundary that would kill them or something like that okay next one yeah okay so then um the fvp coming from this high road or described or explained from this high road then the first thing to bear in mind is that it is a first principles in statistical physics understanding where organisms must maintain their existence and this is what i was referring to earlier on as that first premise so we look at the world we look at how living beings and adaptive systems behave in the environment right and there's one thing that is common to these systems is that they maintain their existence right and this is nice to have as a first premise because it's not um controversial at all i don't think that anybody would um dispute this now the second step of the fep is that how do they maintain their existence how do they maintain this imperative well by minimizing free energy which can be seen or put forward or cast as computational solution to the problem of explaining how the systems maintain their existence so then active inference becomes super relevant here as we know because it is a theory of how living organisms maintain their existence so how they comply with that imperative of remaining alive

maintaining their existence they minimize surprise how via perception and action yeah next slide so then the first premise that we have when we start off our understanding from the fep is that to survive any living organism has to maintain itself in a suitable set of preferable or preferred states while ever avoiding other these preferred states of the environment but then here another very philosophical question comes up for the philosophers in the room which is okay but what are preferable states or prefer states for a certain system how do we define them how do they know right so a bunch of epistemological questions could come up right and uh the fep does reply to that um actually okay so what are these preference states right okay so preferred states are composed of two important aspects so one is those that are defined by niche-specific evolutionary adaptations which for example it would be very unlikely to find as we are doing our research for example to find a butterfly in in a river right because that would not be good for the butterfly so those are niche specific evolutionary adaptations even though there was once a student of mine who told me that some butterflies like to go in the water so maybe this is not a good example i should change this example okay um then the second way of defining preferred states is a very important one which is the learned cognitive goals so as systems interact with the environment of course they learn much more about the environment and themselves and that would be the second way of setting or determining what would be a preferred state in order to maintain existence for an adaptive system okay next one so in order to survive a fish for example has stay in a comfort zone that corresponds to a small subset of all possible states in universe it has to stay in water for example yeah next similarly a human has to ensure that the internal states so for example the physiological variables like body temperature and heart rate remain within acceptable ranges otherwise they will die so or they will become a corpse so then here i find this um extremely um interesting uh for um for for our perspective as human beings uh if we do start with thinking okay human beings need to maintain their body temperature so they need to do that by exerting active control but then there are different ways that we can do that as opposed to non-human animals right so on the one hand we can do and use all the physiology such as sweating but we can also um use some cognitive mechanisms to buy a drink for example uh non-human animals could for example look for water um or we can also have some cultural practices like for example turning on the air conditioning just some examples okay so then um the fvp or um principle can be um thought of as these sort of like steps where we can think about active inference as the biological problem of survival as surprise minimization uh where these surprising states are those that are outside the comfort zone of living organisms so for example a surprising state

for the butterfly would be water and the surprising state for a fish would be being outside of what then the free energy principle i'm sorry frenchy minimization comes very handy because it is a practical and biologically grounded way to explain how adaptive systems minimize the price of these sensory encounters with the world then we can of course as we all know use active inference precisely to explain this coupling uh between a system that is adapting to a continuously changing environment where we can have the system of interest as internal states and the system with which the the and the surrounding environment of the system as external states where as we all know um internal and external states do not directly influence one another but they are indirectly influenced by blanket states such as sensory and active states and the nice thing about it is that we can use the markov blanket formalism that is scale free which interestingly allows us to set up the internal states of scientific interest um for us okay next slide yeah so then these allows us if we want um and note that i am doing literally the higher road to the fep this allows us to now um think about or bring into the conversation if we want the mechanistic predictive processing explanation which is derived from the fep describing cognition as inference that aims to diminish free energy so then what we have is we have this mechanistic explanation of a process theory that allows us to explain uh the activity in the brain in terms of this coupling between internal and external states being mediated by by blankets and sorry by blanket state um meaning uh activity being explained by active inference which then can be very much translated to the language of the pcp and brain hypothesis okay next slide yeah which is uh very of course uh known uh for us um as top-down processes that encode predictions from priors and bottom-up processes that encode sensory observations such as prediction errors so this like two way uh trafficking um occurring uh in brain activity and explaining how brain activity works now um this was the sort of like the high road to the fep and now i want to make like some more highlights some points that i find uh quite relevant um and interesting in in in thinking about things or behavior within the fep framework or by putting in the fep classes so one evening when you do the whole um top-down approach or the high road approach to fep where we start with um that very very simple observation that living beings adaptive systems they want to maintain their existence right even when we start with that and then we find the fep and active inference as a way of explaining how they do that and then we arrive at active inference and can translate all the active inference formalism through to predictive coding and explain how things work in the brain for example it is important to note that even when we do all of that trajectory inference is not necessarily accurate so the optimality of all these mathematics and formalisms is

subjective in the sense that organisms operate on the basis of a subject's um or subjective uh generative model of how the observer observations are generated so i find this picture in the active inference book by pizzulo on and thomas parr and carl um very very uh interesting and useful to precisely convey the point i want to make about the generative model that i feel that sometimes um we just don't um don't see exactly um how it uh really work or is its role is so the genitive model um may be optimized as a new experience is acquired so think about that slide that i have of how does a system how how would we determine what would be the preference stage to a system right so it's precisely through a generative model that allows us to optimize um a model as new experience or new knowledge is required right so then we have um for example this picture here where you see the difference between generative model and a generative process i think i have more things in this slide just yeah okay so then we have this distinction that should be made between a generative model and a generative process and why so well because the genitive model may or may not converge with the generative process and that's sort of what i wanted to highlight now next way yep so the organism the organism's model includes a range of hypotheses about a hidden state which doesn't do not necessarily include the true value of the hidden state of the generative process the generative processes as you see here in the picture the generative process is the world right is how things are in the world right undergenerative model is how i think that things are right so um it includes the way that i think that things are in the world includes a range of hypotheses about the world but those hypotheses do not necessarily include the true value of the world of the generative process right um there's so much many much of it yeah so the models we use to explain our sensorium may include hidden states that do not exist in the outside world and vice versa right so in short what these means is that there does not necessarily need to be a isomorphic match between the generative process and the generative model right so that's the the gist here so this is where action becomes extremely interesting and extremely relevant because action is what is in the middle between genetic model and genetic process right so that's where we get active inference um and what at least i think uh is the the most interesting thing in active inference framework as opposed to other um competing um explanations or frameworks because action is generated on the basis of the inferences made under a generative model right but we just said that the generative model does not necessarily include the true state of affairs of the generative process but notably action is generated on the basis of the inferences made under a generative model next yeah so action is part of the generative process in the sense of is it is through action that we make changes in the world so action is really that interesting thing that

is between the generative model and the generative process action is generated under the generative model which does not necessarily match isomorphically with the generative process and yet action is what is influencing the generative process action is what is influencing the world okay so the the the interesting point here is that despite action being selected from the inferences drawn under a model that is not necessarily accurate as we said it is actually it is this action that is going to affect the generative process so action can be formulated as the minimization of the discrepancy between the generative model and the generative process so it is not only that thing that is sitting in the middle between relative model and generative process but it is actually what can be used and is used to minimize the discrepancy between generative model and so this is this were the points that i uh wanted to make on the fvp and now i want to get us back to think about the fbp under the realism debate and see whether there are some interesting points or things to consider and think about next slide okay so let's think about the free energy principle under the realism debate and let's think about the free energy principle under the option one of realism where the idea was um so so so to say in the definition was that models are accurate representations okay so if we think about um the fup and the realism debate of the first option then we would have to say that generative models are inaccurate representations because they do not necessarily hold accuracy about the generative so we could say that okay so then the generative models are constructed under the fep they are realistic representations with some kind of form of minimal form of realism some minimal form of structuralism and similarity you could say well they are genetic models are representations of genitive processes they're just not accurate they're inaccurate representations and they work so that could be one way um to do that and this is sort of what i think that um kirchoff kiverstein and robertson uh think in their paper but i would be also keen to hear their thoughts about that next yeah so the reason that i say that is because once we look at their paper in the literalist fallacy and fep model building scientific realism and instrumentalism so we saw before in one of the slides when i was presenting realism debate that one way of defining uh realism is that the mappings within the model no yeah the mappings within the model target system so in this case would be the mappings between the fep model allowed to convert truths found in the fvp model into claims about the target system which would be an adaptive system then we find this for example no sorry let me just thank you and then we find for example this passage in the paper similarity we have argued for above between the properties of the genitive model and adaptive behavior such that one can capture real statistical relations in the world which is the genitive process as we just said by

means of generative modeling so it seems that in their paper that's precisely where they're situated in the literature which is precisely on the first name by by realism where there is a similarity between the generative model and the generative process but as we saw just now the generative model does not necessarily need to be accurate about the generative process next one yeah which is precisely what i was saying an organism's uh model includes a range of hypotheses about a hidden state which do not necessarily include the truth value or the true value of the hidden state of the generative process that is not necessarily an isomorphic match of the model of the genetic model and the generative process or of the organisms and the world right okay next so then the second option would be the anti-realist option so what we said about generally speaking we looked at three different ways but what we said was that generative models are empirically adequate so we talked about oh they're viable models um and uh not really true models about um scientific phenomenon of interest they are empirically adequate to the extent and if they allow us to learn more about the target system and if they allow us to make predictions about the target system then they are quite viable they're quite okay so that's the entire list view and in the case of this uh anterior list then representation or matching accuracy is not an issue because models represent in the way that they are used or we use them to form or generate understanding about a target system yeah uh yeah okay so this anti-realism would be um this anti-realism about um fep would be very much in line with for example this brilliant work by albert einstein on mathematics and geometry geometry and experience they'll be perfectly aligned because um what what it says there is that well models are empirically adequate and this is more or less and models will be more or less adequate and um this is important to define what we mean by adequacy right that's the next question there will be more and more or less adequate depending not on accuracy so not on how much the matches the generative process so not on accuracy but on whether or not it minimizes free energy so it is in the sense of whether or not the model is being used allows the minimization of free energy and much less on whether the model being used matches the reality so it's a little bit of a twist there um so then then this non-realist fep uh would also be uh very much in line with for example um complex and dynamical systems theory because um as you know it says organisms as complex systems that persist by seemingly to define the second laws of course they don't really literally defy the second law of thermodynamics but they seem to defy that law of thermodynamics that dictate or prescribe that they should tend to dissipation or chaos but as we observe behavior in the wild and in the world we see that um living organisms adaptive systems they don't do that they don't persist to or they don't succumb to the second

law right so they are extremely complex and dynamic in the sense that their behavior is quite adaptive and that's what makes them so interesting so in this sense if um our fvp models are virtuous not by virtue of them being accurate in the sense that they literally match isomorphically with um genitive process but by virtue of whether or not the model allows the system to minimize free energy then they'll be very much in line with the complex systems theory it would also be i believe in line with an activism theory which i'm not sure if everyone in the room is acquainted with but um it's without there's been some some attempts to make these links between the fep and in activism and um i think that given other work that i've done um in the past um i think that the only viable way of really making this bridge um logically sound is through a non-realist fep a realist fep uh is is much more harder to argue for um uh in their bridge or connection with inactivism so but i'm sure that more most of us are acquainted with it um an activism states that uh uh living beings enact the world to adjust and adapt biologically cognitively and culturally um to the okay um next slide yeah so then this is a very much um it we've covered now the three points i wanted to talk about uh next slate and i just want to conclude by sort of like laying out what the options are for us in this particular uh debate that has become very lively debates in the last one year or so so when we start from the high road and this is important to convey so it's not starting from the low road it's starting from the high road so when we start from the high road to the fvp so when we start from the observation of how biological systems adaptive systems behave and then we have two philosophy of science options which is how do we model these systems and how to think the models by which we explain living adaptive behavior okay so we can take the realist take on it on our models and say well um there is realistic representations our models literally match um isomorphically uh by different uh philosophical strategies or arguments they literally match uh the behavior that we observe and by the same reasoning we can also say that the organisms behaving in the wild they also behave and adopt by virtue of possessing models that literally match the world around them so that's one option the other option is the anterior list option which is to say that the models that we as scientists have the fep models as well as the models that we as living beings have um they allow us to interact with the environment understand the environment because they are empirically adequate not because they accurately represent uh the generative so these are the two options uh that i just wanted to highlight today i think this is my last slide okay thank you so much uh for your time and uh of course i'm very much keen to hear your all your thoughts and questions